

Universitat Politècnica de Catalunya

ESEIAAT

Computational Engineering

Potential flow in a channel

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Contents

1 Problem Definition **3**

1.1 Uniform Potential Flow in a Two-Dimensional Channel 3

1.2 Potential Flow in a Channel Containing a Stationary Cylinder 3

1.3 Potential Flow in a Channel with a Rotating Cylinder 3

2 Theoretical Background **3**

2.1 Formulation of Potential Flow 3

2.2 Stream Function Representation 4

3 Numerical Methodology **4**

4 Code Structure **5**

5 Code Verification **5**

5.1 Validation Case 1: Uniform Potential Flow 5

5.1.1 Analytical Reference Solution 5

5.1.2 Numerical Setup and Boundary Conditions 5

5.1.3 Results and Discussion 6

5.2 Validation Case 2: Flow Around a Stationary Cylinder 6

5.2.1 Analytical Solution 6

5.2.2 Numerical Implementation 6

5.2.3 Numerical Results and Physical Analysis 7

5.3 Validation Case 3: Flow Around a Rotating Cylinder 7

5.3.1 Analytical Model 7

5.3.2 Implementation Details 7

5.3.3 Results and Discussion 7

6 Numerical Analysis **8**

6.1 Grid Convergence Study 8

6.2 Normalized Error as a Function of the Relaxation Parameter δ 8

7 Conclusions **8**

1 Problem Definition

The objective of this study is to simulate an incompressible, inviscid, and steady two-dimensional potential flow within a confined channel, analyzing the influence of a cylindrical obstacle on the velocity and pressure distributions. Both stationary and rotating configurations of the cylinder are examined in order to understand the effect of induced circulation on the flow field.

1.1 Uniform Potential Flow in a Two-Dimensional Channel

This initial configuration consists of a two-dimensional rectangular domain representing a straight channel. The flow is assumed uniform at the inlet with a prescribed constant velocity U in the streamwise direction and no normal velocity component at the upper and lower boundaries, which act as impermeable walls. This case serves as a baseline to validate the numerical solver under simple boundary conditions.

1.2 Potential Flow in a Channel Containing a Stationary Cylinder

In this case, a solid, non-rotating circular cylinder is introduced at the mid-plane of the channel. The cylinder acts as an impermeable obstacle that modifies the potential flow field by imposing a constant stream function value at its surface, which represents the no-penetration condition.

1.3 Potential Flow in a Channel with a Rotating Cylinder

Finally, the case of a rotating cylinder is considered. The angular velocity of the cylinder generates circulation in the surrounding flow field, which induces an asymmetric velocity distribution and produces lift due to the Magnus effect.

2 Theoretical Background

2.1 Formulation of Potential Flow

Potential flow theory describes the motion of an inviscid fluid in regions where viscous effects, heat transfer, and rotational motion are negligible. Under these assumptions, the

stress tensor simplifies as:

$$\tau_{ij} = 0. \tag{1}$$

Additionally, the flow is considered irrotational:

$$\nabla \times \vec{v} = \vec{\omega} = 0, \tag{2}$$

which implies that the velocity field can be expressed as the gradient of a scalar potential function:

$$\vec{v} = \nabla\phi. \tag{3}$$

The velocity potential satisfies Laplace's equation:

$$\nabla^2\phi = 0, \tag{4}$$

whose solution fully characterizes the inviscid flow within the domain, subject to appropriate boundary conditions on solid walls and at the inlet/outlet.

2.2 Stream Function Representation

For two-dimensional, incompressible, and steady flows, it is often convenient to employ the stream function $\psi(x, y)$, defined such that:

$$v_x = \frac{\rho_{ref}}{\rho}\frac{\partial\psi}{\partial y}, \quad v_y = -\frac{\rho_{ref}}{\rho}\frac{\partial\psi}{\partial x}. \tag{5}$$

The continuity equation is automatically satisfied by this definition. Applying the irrotationality condition yields:

$$\frac{\partial}{\partial x}\left(\frac{\rho_{ref}}{\rho}\frac{\partial\psi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\rho_{ref}}{\rho}\frac{\partial\psi}{\partial y}\right) = 0, \tag{6}$$

which constitutes the governing equation for the stream function in a potential flow.

3 Numerical Methodology

The governing Laplace-type equation for the stream function is discretized using a finite volume approach. For each control volume P, Stokes' theorem is applied, yielding the

following flux balance:

$$\Gamma = \oint_C \vec{v} \cdot d\vec{\ell} \approx v_y^e \Delta y_P - v_x^n \Delta x_P - v_y^w \Delta y_P + v_x^s \Delta x_P = 0. \tag{7}$$

Substituting the stream function relations and discretizing in both spatial directions:

$$-\frac{\rho_{ref}}{\rho_e}\frac{\psi_E - \psi_P}{d_{PE}}\Delta y_P - \frac{\rho_{ref}}{\rho_n}\frac{\psi_N - \psi_P}{d_{PN}}\Delta x_P + \frac{\rho_{ref}}{\rho_w}\frac{\psi_P - \psi_W}{d_{PW}}\Delta y_P + \frac{\rho_{ref}}{\rho_s}\frac{\psi_P - \psi_S}{d_{PS}}\Delta x_P = 0. \tag{8}$$

This discretized equation leads to the following algebraic form:

$$a_{P;i,j}\,\psi_{i,j} = a_{E;i,j}\,\psi_{i+1,j} + a_{W;i,j}\,\psi_{i-1,j} + a_{N;i,j}\,\psi_{i,j+1} + a_{S;i,j}\,\psi_{i,j-1} + b_{P;i,j}. \tag{9}$$

The resulting linear system formed by one equation per node is solved iteratively using the Gauss–Seidel algorithm until convergence below a specified residual threshold is achieved.

4 Code Structure

5 Code Verification

5.1 Validation Case 1: Uniform Potential Flow

5.1.1 Analytical Reference Solution

For an inviscid and irrotational flow with uniform inlet velocity U, the analytical stream function reads:

$$\psi(x, y) = -Vx + Uy, \tag{10}$$

which represents a purely horizontal flow. This case is used to verify the solver's ability to reproduce a constant velocity field.

5.1.2 Numerical Setup and Boundary Conditions

The computational domain has dimensions $L \times H = 5 \times 1\,\text{m}$. Boundary conditions are imposed as follows:

- **Inlet:** $\psi(0, y) = Uy$.
- **Outlet:** $\frac{\partial \psi}{\partial x} = 0$.
- **Walls:** $\psi = 0$ at $y = 0$ and $y = H$.

A uniform inlet velocity of $U = 1\,\mathrm{m/s}$, $T = 300\,\mathrm{K}$, and $p = 101{,}325\,\mathrm{Pa}$ is prescribed. The numerical mesh consists of $N = M = 20$ divisions.

5.1.3 Results and Discussion

The computed streamlines show uniform spacing and directionality along the channel. The horizontal velocity component v_x remains approximately constant throughout the domain, while $v_y \approx 0$, confirming the one-dimensional character of the flow. Temperature and pressure fields remain nearly uniform, as expected for a potential, incompressible flow.

5.2 Validation Case 2: Flow Around a Stationary Cylinder

5.2.1 Analytical Solution

For an ideal potential flow around a stationary cylinder of radius R centered at (x_0, y_0), the stream function is given by:

$$\psi(x, y) = -Vx + Uy + \frac{\mu}{2\pi} \frac{(y - y_0)\cos\alpha - (x - x_0)\sin\alpha}{(x - x_0)^2 + (y - y_0)^2}, \tag{11}$$

representing the superposition of a uniform flow and a dipole potential.

5.2.2 Numerical Implementation

A structured Cartesian mesh is used to discretize the channel ($L \times H = 2 \times 1\,\mathrm{m}$). Each grid node (x_P, y_P) is evaluated to determine whether it lies within the cylinder:

$$r_P = \sqrt{(x_P - x_c)^2 + (y_P - y_c)^2}. \tag{12}$$

Nodes satisfying $r_P < R$ are assigned a constant stream function value $\psi_P = \psi_{body} = v_{in}(H/2)$, ensuring impermeability. The coefficients of solid elements are modified as:

$$a_P = 1, \quad a_E = a_W = a_N = a_S = 0, \quad b_P = \psi_{body}. \tag{13}$$

This treatment ensures that solid nodes do not contribute to the external flow solution.

5.2.3 Numerical Results and Physical Analysis

The velocity field exhibits stagnation points at the front and rear of the cylinder, where the local velocity drops to zero, and maximum velocity along the lateral sides, reaching approximately twice the freestream value. The flow symmetry with respect to the horizontal axis confirms the irrotational nature of the solution. Temperature and pressure variations remain minimal due to the inviscid assumption.

In accordance with d'Alembert's paradox, both lift and drag coefficients approach zero:

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$$C_L \approx 0, \quad C_D \approx 0. \tag{14}$$

5.3 Validation Case 3: Flow Around a Rotating Cylinder

5.3.1 Analytical Model

When the cylinder rotates with angular velocity ω, the total stream function becomes:

$$\psi(x, y) = \psi_{\text{static}} - \frac{\Gamma}{2\pi} \ln\left(\sqrt{(x - x_0)^2 + (y - y_0)^2}\right), \tag{15}$$

where the circulation $\Gamma = 2\pi R^2 \omega$.

5.3.2 Implementation Details

The value of the stream function inside the rotating body is modified to include the rotational contribution:

$$\psi_{body} = v_{in}\left(\frac{H}{2}\right) + R_{cyl}^2 \omega. \tag{16}$$

The numerical scheme remains identical to that of the stationary cylinder, except for this modification.

5.3.3 Results and Discussion

Rotation introduces a distinct asymmetry in the flow field. The stagnation points shift along the cylinder surface, and velocity increases on the side where the rotational motion aligns with the freestream velocity. Pressure decreases in these high-speed regions, giving

rise to a net lift force consistent with the Magnus effect:

$$C_L \approx -\frac{\Gamma}{v_\infty R_{cyl}}, \qquad C_D \approx 0. \tag{17}$$

This confirms that rotation induces circulation and consequently lift, while drag remains negligible in potential flow.

6 Numerical Analysis

6.1 Grid Convergence Study

To evaluate numerical accuracy, the velocity field computed around a stationary cylinder is compared with the analytical solution for increasing mesh resolutions. The pointwise difference between both fields defines the local error, which is filtered to remove outliers. As shown by the trend, the global error decreases monotonically with finer mesh discretization, confirming the consistency and convergence of the implemented method.

6.2 Normalized Error as a Function of the Relaxation Parameter δ

The normalized error of the velocity magnitude v_P is analyzed as a function of the convergence parameter δ for different grid densities. Each error curve is normalized by its maximum to facilitate comparison. Results show that smaller δ values yield higher accuracy, though with diminishing returns beyond a certain threshold. Finer meshes consistently exhibit lower errors across all δ, confirming the numerical stability of the method.

7 Conclusions

A numerical model for two-dimensional potential flow in a channel has been successfully implemented and validated. The solver accurately reproduces theoretical results for uniform flow, flow around a stationary cylinder, and flow around a rotating cylinder. The inclusion of rotation correctly induces circulation and lift, consistent with the Magnus effect, while drag remains negligible.

Mesh refinement studies confirm numerical convergence, and the method demonstrates robustness and reliability for analyzing inviscid, incompressible flows around bluff bodies.

The developed framework can serve as a foundation for more complex simulations, such as compressible or viscous flow extensions.